

15.S06: AI and Machine Learning Research in Finance

SUGGESTED PROJECTS

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Instructions

- Please form a team of four students. You can view the class roster on Canvas.
- Read the project descriptions below and discuss with your team which projects you would like to pursue.
- After the first lecture, specify the top three project choices for your team using the sign-up sheet (to be posted on Canvas) by Thursday, September 10.
- Each team will be assigned one project and a research advisor by Friday, September 11.
- Teams are expected to do the following:
 1. Meet with your advisor weekly throughout the semester to discuss progress and next steps;
 2. Present a preliminary research plan to the class (Week 2);
 3. Present the most relevant paper(s) to the class (Weeks 3-4). Explain their methodology, main results, contributions to the literature, limitations, and potential extensions;
 4. Provide project updates to the class (Weeks 6-7 and Weeks 10-11);
 5. Deliver a final presentation (Week 15).

1 Prediction Markets as Macro Data: Kalshi and Polymarket

Goal

Event markets on macroeconomic outcomes (CPI releases, Fed decisions, elections) have become liquid enough to take seriously as a source of real-time probabilistic forecasts. A new literature studies their informational content and microstructure [1, 2, 3]. Students will assemble a dataset of contract prices from Kalshi (REST API) and/or Polymarket (fully public on-chain trades) and evaluate these markets as forecasting devices and as trading venues.

Research Questions

1. Calibration (are Kalshi expectations rational?): do market prices on macro events beat professional forecasters (SPF) and option-implied distributions? Is there a favorite-longshot bias?
2. Price discovery: how fast do event prices incorporate scheduled releases and Fed communication? Compare against high-frequency moves in fed funds futures.
3. Leverage Kalshi’s data to predict macroeconomic quantities, such as unemployment, GDP growth, and CPI. Advanced ML techniques, such as economic transformers, may be used here.

Data

- Kalshi API (historical prices and order books); Polymarket on-chain data.
- SPF consensus forecasts; fed funds futures; OptionMetrics for implied distributions; stock indices from WRDS.

References

- [1] Diercks, A., Katz, J., & Wright, J. (2026). “Kalshi and the Rise of Macro Markets.” NBER Working Paper.
- [2] Bürgi, C., Deng, W., & Whelan, K. (2026). “Makers and Takers: The Economics of the Kalshi Prediction Market.” Working Paper.
- [3] Bartlett, R., & O’Hara, M. (2026). “Adverse Selection in Prediction Markets: Evidence from Kalshi.” Working Paper.
- [4] Wolfers, J., & Zitzewitz, E. (2004). “Prediction Markets.” *Journal of Economic Perspectives*, 18(2).
- [5] Kelly, B., Kuznetsov, B., Malamud, S., & Xu, T. (2026). “Artificial Intelligence Asset-Pricing Models.” SSRN Working Paper.

2 Predicting the Valuation of Non-Listed Firms

Goal

This project is about leveraging ML techniques to improve the corporate valuation of non-listed firms (a topic of great interest to private equity investors and VC funds; see Bonelli [1]). Hommel, Landier, and Thesmar [2] bring valuation theory to the data: using public firms, they find that “predicted IRRs” based on firm characteristics provide the best way to discount future cash flows, yielding estimates closest to observed market values. This project applies this insight to privately held firms. Using deal-level data on private equity and venture capital transactions, students will apply modern ML techniques and enrich the data with newspaper text to build the strongest private-firm valuation model possible.

Research Questions

1. Benchmarks: on a sample of PE transactions with observed valuations, how accurate are the standard methods — peer multiples (EV/EBITDA, EV/Sales from public comparables), and DCF variants in the spirit of [1]? Use median absolute pricing error as the yardstick, and document where each method fails (small firms, unprofitable firms, hot sectors).
2. Students may also develop a framework similar to that of Hommel, Landier, and Thesmar [2] to predict companies’ future cash flows based on their characteristics. They can use these predicted cash flows to compute internal rates of return and apply a similar methodology to private transactions.
3. Machine learning: train flexible models (gradient boosting, neural nets) on fundamentals, deal characteristics, sector, and market conditions to predict transaction valuations, in the spirit of ML-based relative valuation for public firms [3]. How large is the gain over multiples? Which features matter (SHAP values), and does the model effectively learn time-varying, sector-specific multiples?
4. Text enrichment: match target companies to newspaper and trade-press coverage, and use LLMs to extract what financials cannot show — growth narrative, competitive position, management quality, controversy. Does text improve valuation accuracy, and is the gain largest where financial data is sparsest (early-stage and private firms)?

Data

- Private-equity and VC transaction data: PitchBook and Preqin, available through WRDS.
- Public comparables: Compustat and CRSP for multiples and market conditions.
- Text: newspaper and trade-press articles matched to target firms (Factiva / ProQuest; Raven-Pack for larger targets).

- Coverage of private-firm financials is incomplete, and part of the project is to build a pipeline that fills gaps in the transaction data for private firms used to train the ML model.

References

- [1] Bonelli, M. (2026). “Data-Driven Investors.” *The Review of Financial Studies*, forthcoming.
- [2] Hommel, N., Landier, A., & Thesmar, D. (2023). “Corporate Valuation: An Empirical Comparison of Discounting Methods.” NBER Working Paper No. 30898.
- [3] Gornall, W., & Strebulaev, I. (2020). “Squaring Venture Capital Valuations with Reality.” *Journal of Financial Economics*, 135(1).
- [4] Geertsema, P., & Lu, H. (2023). “Relative Valuation with Machine Learning.” *Journal of Accounting Research*, 61(1).
- [5] Kaplan, S., & Ruback, R. (1995). “The Valuation of Cash Flow Forecasts: An Empirical Analysis.” *Journal of Finance*, 50(4).

3 Long-Term News and Index Returns

Goal

Hirshleifer, Mai, and Pukthuanthong [1] extract discourse topics from more than a century of *New York Times* coverage and show that “war discourse” is priced in the cross section of U.S. stock returns; companion work [2] shows that the same discourse measure predicts the equity premium in the time series over 160 years. This project is based on their framework. Students will first replicate the original results using the authors’ replication code and topic model, then replace traditional NLP (LDA-family topic models) with LLM prompting. The next step is to extend the analysis of these discourse measures to *index* returns in the U.S. and internationally.

Research Questions

1. Replication: using the authors’ package, reproduce the construction of the war-discourse factor and its main asset-pricing results (priced war beta in the cross section; time-series predictability of the market return).
2. Measurement: replace the topic model with LLM-based extraction — prompt an LLM to score each front page or article for war intensity, the threat of disaster, and related narratives (an agentic, multi-prompt approach may be needed at this scale, with careful attention to cost, sampling, and prompt stability). Do LLM-based discourse series track the LDA-based ones? Where they diverge, which measure better predicts returns?
3. Students may explore other geopolitical or economic events.

Data

- Long-term historical text: ProQuest Historical Newspapers (*New York Times*, *Wall Street Journal*, and others, back to the 1850s), available through MIT Libraries: <https://libguides.mit.edu/c.php?g=1253643&p=9182104#s-lg-box-wrapper-34468150>. The authors’ replication package provides the topic-model pipeline.
- Index returns: World Indices by WRDS (country and regional index returns); CRSP value-weighted market and individual stock returns for the replication step; Goyal–Welch predictor data for benchmarking.

References

- [1] Hirshleifer, D., Mai, D., & Pukthuanthong, K. (2025). “War Discourse and the Cross Section of Expected Stock Returns.” *Journal of Finance*, 80(6): 3589–3637.
- [2] Mai, D., & Pukthuanthong, K. (2024). “War Discourse and Disaster Premium: 160 Years of Evidence from the Stock Market.” Working Paper.

- [3] Manela, A., & Moreira, A. (2017). “News Implied Volatility and Disaster Concerns.” *Journal of Financial Economics*, 123(1): 137–162.
- [4] Bybee, L., Kelly, B., Manela, A., & Xiu, D. (2024). “Business News and Business Cycles.” *Journal of Finance*, 79(5).
- [5] Goyal, A., & Welch, I. (2008). “A Comprehensive Look at the Empirical Performance of Equity Premium Prediction.” *Review of Financial Studies*, 21(4).

4 Taming the Covariance Matrix: Shrinkage for Better Portfolios

Goal and Research Objective

Estimating the covariance matrix of returns is difficult when the number of assets is large relative to the available time series, and small estimation errors can be greatly amplified by portfolio optimization. The project will replicate the nonlinear shrinkage methods in [2] and the Universal Portfolio Shrinkage Approximator (UPSA) in [1], determine when they improve out-of-sample portfolio performance, and develop an economically motivated extension. Before beginning the empirical work, the team should formulate a precise research objective that identifies the portfolio problem, benchmark estimators, and proposed extension.

Research Questions

1. Can we reproduce the simulations and main empirical portfolio results in [1]? Compare UPSA with the sample covariance matrix, linear and nonlinear shrinkage, factor-model estimators, and simple portfolio benchmarks.
2. Where do the gains come from? Study how the methods treat different principal components of returns and how performance varies with the number of assets, estimation-window length, portfolio objective, and market conditions.
3. Can the estimator be improved for an economically important setting? Possible directions include time-varying covariance matrices, heavy tails, transaction costs and turnover, gross-exposure or short-sale constraints, alternative shrinkage targets, and portfolio objectives other than minimum variance.

Data and Evaluation

- Begin with the authors' replication data and anomaly portfolios. Extend the analysis to CRSP stocks or another large asset universe.
- Use rolling or expanding estimation windows. Evaluate realized volatility, Sharpe ratio, draw-down, turnover, and performance net of trading costs. All tuning choices must be made without using the final test sample.

References

- [1] Kelly, B.T., Malamud, S., Pourmohammadi, M., & Trojani, F. (2026). "Universal Portfolio Shrinkage." *Review of Financial Studies*, forthcoming.
- [2] Ledoit, O., & Wolf, M. (2017). "Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks." *Review of Financial Studies*, 30(12): 4349–4388.
- [3] DeMiguel, V., Garlappi, L., & Uppal, R. (2009). "Optimal Versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?" *Review of Financial Studies*, 22(5): 1915–1953.

5 Which Expert, When? Alpha Models Across Market Regimes

Goal and Research Objective

Financial predictors are unstable: a signal that works in one market environment may fail or reverse in another. This project asks whether an economically structured mixture-of-experts (MoE) model can adapt to this instability. Students will organize signals into economic “domain experts” and use a gating model to combine or select them based on current market conditions. Before building the model, the team should formulate a precise research objective that specifies the prediction target, expert definitions, gating information, and benchmark model-selection approach.

Research Questions

1. Does grouping signals by economic category produce experts that genuinely specialize across assets, horizons, or market regimes, relative to a single pooled alpha model and simple forecast averaging?
2. Can a soft or sparse MoE gate use ex ante macroeconomic, volatility, liquidity, and valuation states to select experts out of sample? Are the learned routing weights stable and economically interpretable?
3. How does the MoE compare with Bayesian model selection/averaging and Markov regime-switching models in predictive accuracy, calibration, portfolio performance, turnover, and robustness to structural change?

Data and Design

- CRSP and Compustat for returns and firm characteristics; standard macro-financial state variables from FRED, option-implied volatility, and aggregate valuation and liquidity measures.
- The [Jensen–Kelly–Pedersen Global Factor Data](#) groups 153 factors into 13 themes across 93 countries and is well suited to defining experts and studying global regimes.
- Use chronological, nested splits. Compare hard and soft routing with pooled and separate models, equal-weighted forecasts, Bayesian model averaging, and a regime-switching benchmark. Report statistical accuracy and performance after trading costs.

References

- [1] Jacobs, R.A., Jordan, M.I., Nowlan, S.J., & Hinton, G.E. (1991). “Adaptive Mixtures of Local Experts.” *Neural Computation*, 3(1): 79–87.
- [2] Ang, A., & Timmermann, A. (2012). “Regime Changes and Financial Markets.” *Review of Financial Studies*, 25(4): 1017–1050.
- [3] Avramov, D. (2002). “Stock Return Predictability and Model Uncertainty.” *Journal of Financial Economics*, 64(3): 423–458.

6 Can AI Read Earnings? Building an AI Equity Analyst

Goal and Research Objective

Build an AI equity analyst that predicts how stock prices respond to earnings announcements. The project will begin by replicating the optimized agentic systems and evaluation framework in [1] and using historical data for backtesting. The team will then design and test improvements and sign up for the Q4 2026 round of the authors’ live [Explaining Markets](#) competition. Before building the system, the team should formulate a precise research objective that identifies the prediction target and the agent component or economic hypothesis to be tested, beyond simply improving leaderboard performance.

Research Questions

1. Can we reproduce the historical benchmark in [1], including the incremental explanatory power of the AI prediction beyond the earnings-surprise benchmark?
2. Which components create value: retrieval of filings and prior calls, structured financial analysis, tools for numerical calculations, multiple specialized agents, debate or critique, model and prompt choice, or forecast aggregation?
3. What does the analyst learn about market responses? Identify the qualitative information and interactions that improve predictions, and test whether performance varies by firm type, information environment, announcement content, and market conditions.

Data and Evaluation

- Use the competition SDK, API, and historical event data for backtesting, supplemented where useful with SEC filings, earnings-call transcripts, I/B/E/S, CRSP, Compustat, RavenPack, and Capital IQ.
- Respect the information cutoff for each historical and live event. Freeze prompts and model choices before the Q4 live evaluation, log all inputs and outputs, and compare against simple earnings-surprise, text-sentiment, and non-agentic LLM baselines. Evaluate out-of-sample R^2 , calibration, rank accuracy, cost, and robustness across events.

References

- [1] Koijen, R.S.J., & Levy, B. (2026). “Assessing the Benefits of Optimized Agentic AI Systems for Asset Pricing.” Working Paper.
- [2] Tetlock, P.C. (2007). “Giving Content to Investor Sentiment: The Role of Media in the Stock Market.” *Journal of Finance*, 62(3): 1139–1168.

- [3] Lopez-Lira, A., & Tang, Y. (2023). “Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models.” Working Paper.
- [4] Graham, J., Harvey, G., & Jha, M. (2026). “CFOs Meet LLMs.” SSRN Working Paper.

7 The “Virtue” of Complexity: Signal or Mirage?

Goal and Research Objective

Does an extremely overparameterized model uncover genuine return predictability, or can its apparent success be an artifact of implementation choices and a favorable historical episode? [1] argue that return prediction and portfolio performance can improve as model complexity grows far beyond the sample size. [2] and [3] offer distinct critiques of the mechanism and empirical evidence. The project will reconstruct this debate, replicate the competing results, and then seek evidence for or against the virtue of complexity in one carefully chosen market setting. The team should formulate a precise research objective that states which mechanism in the debate will be tested and in which market setting.

Research Questions

1. Can we reproduce the complexity-performance curves in [1] and the main diagnostic results in [2] and [3] using a common code base and identical samples?
2. Which mechanisms explain the results: recovery of a genuinely high-dimensional signal, implicit recency- and volatility-timed momentum, the zero-intercept restriction, the aggregation of performance across models, or other tuning and implementation choices?
3. Does complexity help in another market? Choose one setting—for example, international equities, corporate bonds, currencies, commodities, options, or earnings-announcement returns—and pre-specify the prediction target, information set, benchmarks, and evaluation design.

Design and Evaluation

- Trace the algebra and simulation evidence in all three papers before running the empirical horse race. Vary model complexity, training-window length, intercept treatment, regularization, feature volatility, and the persistence or reversal of returns.
- Use nested rolling validation and untouched test periods. Report out-of-sample R^2 , forecast decomposition, Sharpe ratio, drawdown, turnover, and trading-cost sensitivity. Diagnose which observations, regimes, and implicit strategies account for any gains.

References

- [1] Kelly, B.T., Malamud, S., & Zhou, K. (2024). “The Virtue of Complexity in Return Prediction.” *Journal of Finance*, 79(1): 459–503.
- [2] Nagel, S. (2025). “Seemingly Virtuous Complexity in Return Prediction.” Becker Friedman Institute Working Paper No. 2025-104.
- [3] Buncic, D. (2026). “Simplified: A Closer Look at the Virtue of Complexity in Return Prediction.” Working Paper, revised March 2026.